

Chapter 4 – Functions with Return Values

SLIDE 1 — What Is a Return Value?

Definition

A return value is the result sent back by a function after completing its task.

A function can:

- Receive values through parameters.
- Perform a calculation.
- Return the result to the calling part of the program.

Analogy

Think of an ATM:

- You request money.
- The ATM processes your request.
- The ATM returns the money to you.

Input → Processing → Return Value

Small Example

None

```
int add() {  
    return 5 + 3;  
}
```

The function returns the value 8.

Remember

A return value is the result sent back by a function.

SLIDE 2 — Why Are Return Values Used?

Why do we use return values? 🤔

Return values are used to:

1. Send results back to the calling function.
2. Store the result in a variable.
3. Reuse the result in another calculation.
4. Make functions flexible.
5. Separate calculation from output.
6. Improve code readability.
7. Use functions in conditions.

Important Statement ★

A function with a return value gives a result instead of directly displaying it.

Base → Professional

Base Level:

```
None
int getNumber() {
    return 10;
}
```

Professional Level:

```
None
int calculateTotal(int price, int quantity) {
    return price * quantity;
}
```

The second function accepts inputs and returns a useful result.

Return vs Print

return

printf()

Sends a value back Displays a value

Used to give a result Used to show output

Can store the result Prints directly on screen

SLIDE 3 — Return Types

What is a Return Type? 🧠

The return type tells the compiler what type of value a function will return.

Common Return Types

Return Type	Meaning
<code>int</code>	Returns a whole number
<code>float</code>	Returns a decimal number
<code>char</code>	Returns a character
<code>void</code>	Returns no value

Example

None

```
int square(int n) {  
    return n * n;  
}
```

Here:

- `int` → Return type
- `square` → Function name
- `n` → Parameter
- `return n * n;` → Sends back the result

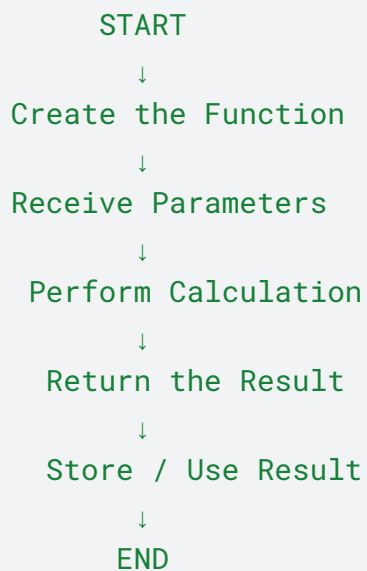
Important Statement

The returned value should match the function's return type.

SLIDE 4 — Flowchart: Function with Return Value

Flowchart

None



Example

None

```
#include <stdio.h>

int add(int a, int b) {
    return a + b;
}

int main() {
    int result = add(4, 6);
    printf("%d", result);
    return 0;
}
```

Output

None

10

Explanation

1. `add()` receives 4 and 6.
2. The function calculates `4 + 6`.
3. `return` sends back 10.
4. The value is stored in `result`.
5. `printf()` displays the result.

Remember ★

Return gives the value back; `printf()` shows it.

SLIDE 5 — Witty Recap

The Function Story 😊

Imagine ordering food from a restaurant:

1. You give the order → Parameters
2. The chef prepares the food → Function execution
3. The chef gives the food back → Return value

4. You receive and use it → Store or display the result

Memory Trick 🧠

Parameters go in. Return values come out.

Final Recap ★

- A return value is the result sent back by a function.
- `return` is used to send the result.
- `int`, `float`, and `char` can be return types.
- `void` means no value is returned.
- A returned value can be stored in a variable.
- `return` and `printf()` have different purposes.

Final Formula

Input → Process → Return → Store / Display

Questions

1. Theory MCQ

What does a function with a return value do?

A. Only displays text B. Sends a result back C. Stops the program D. Deletes a variable

💡 Hint: The function gives a result to the caller.

📖 Glossary: Result — The value produced by a function.

Answer: B. Sends a result back

2. Output Guess

None

```
#include <stdio.h>
```

```
int square(int n) {  
    return n * n;  
}
```

```
int main() {  
    printf("%d", square(5));  
    return 0;  
}
```

What is the output?

💡 Hint: Multiply 5 by itself.

📖 Glossary: Square — Multiplying a number by itself.

Answer:

None

25

3. Fill the Missing Code

None

```
#include <stdio.h>
```

```
int doubleNumber(int n) {  
    _____ n * 2;  
}
```

💡 Hint: Use the keyword that sends the answer back.

📖 Glossary: Return – Sends a value back from a function.

Answer: return

4. Jumbled Sentence —

Arrange the words correctly:

function / call / the / To / execute / code

💡 Hint: Start with “To”.

📖 Glossary: Execute — To run the code.

Answer: To execute the code, call the function.

5. Fill the Missing Code –

None

```
#include <stdio.h>

void _____() {
    printf("Welcome");
}

int main() {
    greet();
    return 0;
}
```

💡 Hint: The function name must match the name used in the call.

📖 Glossary: Function name – The name used to identify and call a function.

Answer: greet